

Arunabha Majumder

Mechanical Designer | 3D Design, Strength Calculation (FEA), Drawings & Prototyping

+45 71 62 24 00 • Aalborg, Denmark • somrkmv1997@gmail.com • 97arunabhasit027@gmail.com

LinkedIn: [linkedin.com/in/arunabha-majumder-681264107](https://www.linkedin.com/in/arunabha-majumder-681264107) • Portfolio: arunabha22.github.io/arunabha-majumder

SUMMARY

Mechanical designer with 3+ years of hands-on 3D design, strength calculation (FEA) and prototype building. Takes projects from concept and detail design to workshop drawings, manufacture and testing, and has built machines outside work since student years, including a Go-Kart from the ground up.

KEY RESULTS

- **Concept to working hardware:** designed, drew, built and tested a load-carrying actuated device (Best Paper Award, 2026)
- **>25%** energy reduction through a better technical solution
- **<10%** error on a validated measurement method

EXPERIENCE

Research Engineer – Mechanical Design & Actuator Development (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Delivered a working prototype from concept to tested hardware by taking ownership of the design project: concept design, 3D modelling and detail design in SolidWorks, workshop drawings for machined and 3D-printed parts, and hands-on assembly and testing.
- Verified the structural safety and weight of load-carrying parts before manufacture by using strength calculation with FEA for stiffness, strength and weight under expected load cases, then iterated the design for manufacturability.
- Cut energy consumption by more than 25% in prototype testing (Best Paper Award, IFToMM ISRM 2026) by defining the technical solution for a hybrid actuation system: a parallel spring working with a 6 Nm motor.
- Achieved force estimation accurate to below 10% RMS error, validated against a force sensor, by developing a sensorless method for a 5-bar planar parallel robot using variable-stiffness actuators and the robot Jacobian; applied it to measure human arm impedance in the gait lab.
- Reduced user muscle effort by more than 15%, measured with EMG, by implementing an Assist-as-Needed controller on the prototype; ran data collection for a 12-participant study in a multidisciplinary team.
- Mentored and supervised Bachelor's (B.Tech) students on engineering projects (2023 – 2025).

MTech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Built an empirical actuator model from test data by fabricating pneumatic artificial muscles in-house from raw materials and setting up a test rig with valves, pressure sensors and DAQ.
- Reduced position-tracking error to 0.3–0.78%, outperforming classical PID under varying load, by designing a neural-network-based controller.

HANDS-ON ENGINEERING

- **Go-Kart (2017):** Designed, fabricated and assembled a custom Go-Kart from the ground up.
- **Efficycle (2018):** Built a human-electric hybrid vehicle in a multidisciplinary team for the Efficycle competition.
- **Rehabilitation device (2018–2019):** Led CAD design, fabrication and user testing of an upper-limb rehabilitation exoskeleton; Best Major Project Award and INR 100,000 TATA Technologies Innovation Grant.

- **Workshop:** Fabricated actuators from raw materials; machining, 3D printing and assembly for every prototype.

SKILLS

- **Strength calculation & FEA:** Stiffness, strength, load cases, weight optimisation
- **3D design & drawings:** SolidWorks (CSWA), Fusion 360, detail design, workshop drawings, assemblies, DFA
- **Manufacturing & prototyping:** Machining, 3D printing, assembly, test rigs, DAQ
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

REFERENCES

Available upon request.